Multi-Agent Debate for LLM Evaluation: A Survey

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Abstract

While large language models (LLMs) have revolutionized text generation, there is still no system that robustly evaluates output quality without relying on human judgment. In light of this, the Multi-Agent Debate Evaluation Framework (MADEF), where LLMs debate to produce a final judgment on AI-generated data, has gained increasing interest. We survey this emerging area by introducing a comprehensive analysis along three dimensions: dataset, framework design, and evaluation rubrics. Next, we present results across three domains using four state-of-the-art MADEFs. Finally, we discuss the strengths and limitations of each framework, highlight key challenges, and provide recommendations for future work.

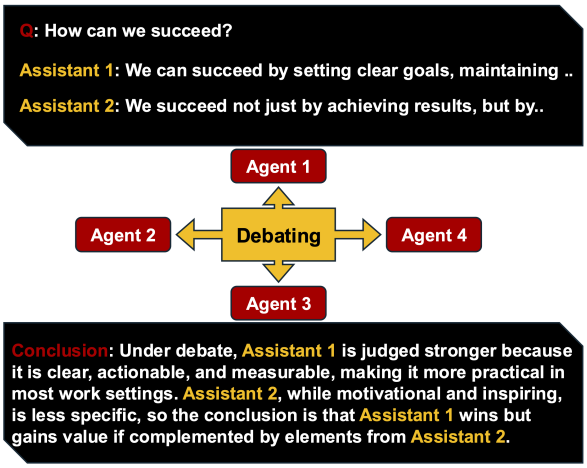

Figure 1: Diagram of MADEF: Given AI-generated text, multiple agents `debate`, leading to a final judgment.

1 Introduction

LLMs have demonstrated remarkable capabilities in generating coherent and contextually rich text across tasks such as reasoning, dialogue, and summarization. However, evaluating the quality and correctness of LLM-generated outputs remains a major challenge. Existing n-gram automatic metrics such as BLEU-4 (Papineni et al., 2002), ROUGE-L (Lin, 2004), and embedding-based metrics such as BERTScore (Zhang et al., 2020) often fail to capture semantic nuance, factual accuracy, or domain-specific reliability. Meanwhile, human evaluation is a costly, time-consuming, and difficult-to-scale alternative (Gao et al., 2025).

In light of this, recent studies explore **LLM-as-a-judge**, where a single LLM is prompted to directly assess or compare model-generated outputs without reference answers. These *single-agent* evaluation systems offer scalability and flexibility, but also introduce several limitations: the judge can hallucinate, produce inconsistent scores in multiple prompts, or reinforce its own biases without external challenge (Szymanski et al., 2024).

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To address these issues, there is growing interest in **Multi-Agent Debate Evaluation Frameworks (MADEF)**, in which multiple LLMs engage in structured debate (critiquing, defending, and judging generated texts) to arrive at a final, self-consistent decision (Oriol et al., 2025). MADEF aims to improve robustness, mitigate single-judge bias, and enhance interpretability by exposing intermediate reasoning traces. However, despite promising early results, it remains unclear which MADEF variant provides the most reliable evaluation across different domains and task types. Moreover, MADEFs introduce new challenges: they require multiple interacting LLMs, increasing inference cost and raising open questions about how to design computationally efficient configurations.

In this study, we conduct a comprehensive analysis of representative MADEFs, examining their performance and adaptability across diverse datasets. Specifically, we evaluate four MADEFs: (1) Debatable-Intelligence (Sternlicht et al., 2025), (2) MORE (Bandi and Harrasse, 2024), (3) DEBATE (Kim et al., 2024), and (4) ChatEval (Chan et al., 2023) over three domains: (1) open-domain

QA and dialogue, (2) multi-step mathematical reasoning, and (3) medical question-answering. We also analyze human–MADEF alignment to assess whether these frameworks replicate human judgment patterns, and we examine debate histories to understand how final decisions are formed. To guide our analysis, we pose two research questions:

- **RQ1:** For a given dataset domain, does a specific MADEF achieve superior performance?
  H1: Domain-specialized MADEFs will outperform others on tasks aligned with their design assumptions.
- **RQ2:** Must an effective MADEF be complex to implement or computationally expensive?
  H2: Lightweight, lower-cost MADEF configurations can achieve performance comparable to resource-intensive frameworks.

Through this work, we aim to clarify the trade-offs among existing MADEFs and establish a foundation for developing more efficient and transparent LLM-as-a-judge paradigms. Our results show that MADEFs can approximate human judgment across multiple domains while varying substantially in cost, interpretability, and robustness. Notably, increased complexity does not necessarily yield better evaluation quality.

2 Related Work

2.1 Single-Agent Evaluation Frameworks

Single-agent evaluation frameworks rely on a *single* LLM to assess or compare model outputs. These systems typically prompt the model to provide a score or preference judgment.

GPTScore Fu et al. (2023) propose a general LLM-as-a-judge framework that evaluates generation quality using task-specific natural-language rubrics instead of relying on surface-form similarity metrics. Given a prompt, reference, and system output, GPTScore prompts a strong LLM to directly assign a scalar quality score, and demonstrates high correlation with human judgments across summarization, dialogue, and data-to-text tasks.

G-Eval Liu et al. (2023) introduce an LLM-based evaluation framework that improves scoring consistency by combining structured scoring rubrics with Chain-of-Thought style reasoning. Instead of asking the model to output only a final score, G-Eval guides the model through step-by-step justification followed by a calibrated numeric rating, which empirically yields higher inter-annotator agreement with humans.

MADEF	Tasks	Debaters
Deb.-Int.	debate speeches	Aff. & Neg. debaters & n judges
MORE	multi-turn dialogue	2 advocates & 1 judge
DEBATE	summarization	1 critic & 1 scorer
ChatEval	open-ended QA	1 Public & 1 critic
Single.	task-agnostic	N/A

Table 1: Overview of MADEFs. Deb.-Int. denotes Debatable Intelligence. Single. denotes a single-agent evaluation framework.

SelfCheckGPT Manakul et al. (2023) focuses on hallucination detection rather than scalar quality scoring. It assumes black-box access and probes factual consistency by sampling multiple continuations of the same claim and measuring self-agreement. Low consistency signals hallucination, enabling zero-resource detection without external knowledge or labels and offering a complementary evaluation axis to rubric-based LLM-as-a-judge methods.

Although scalable and easy to deploy, single-agent methods suffer from well-known limitations: they may hallucinate, produce inconsistent scores across prompts, and reinforce internal biases due to lack of external challenge. These shortcomings have motivated the exploration of multi-agent evaluation settings.

2.2 MADEF

We illustrate MADEF in Figure 1, where multiple LLM agents debate over AI-generated text to produce a final judgment that is often more reliable than a single-judge evaluation. Research on MADEF spans diverse domains, summarized in Table 1. We categorize existing frameworks into two areas: (1) mathematical and scientific reasoning, and (2) open-domain QA and dialogue. This organization highlights how different MADEFs adapt to domain-specific evaluation needs.

2.2.1 Mathematics & Scientific Reasoning

Reasoning-intensive domains require evaluation frameworks that handle logical validity, precision, and specialized criteria (Lu et al., 2024).

Debatable Intelligence Sternlicht et al. (2025) evaluate nuanced attributes such as argument strength, coherence, and organization by prompting LLMs to score debate speeches across multiple

qualitative dimensions. Using over 600 human-annotated debate speeches (Slonim et al., 2021), they report agreement and correlation with human scores as well as behavioral consistency analyses. While advanced LLMs occasionally approximate individual human judgments, results show that divergence remains common, especially in subtle or context-dependent cases.

2.2.2 Open-domain QA & Dialogue

These tasks pose additional challenges due to open-endedness and multidimensional evaluation criteria (Huang et al., 2020).

MORE Bandi and Harrasse (2024) introduce a courtroom-inspired evaluation framework in which agents defend and critique responses. They propose two architectures: Multi-Advocate One-Round Evaluation (MORE) and Single Advocate Multi-Round Evaluation (SAMRE), that facilitate structured adversarial interactions. Evaluated on MT-Bench (Zheng et al., 2023a), containing 80 questions and 3,300 expert preference pairs, both configurations outperform a single-judge baseline, with SAMRE achieving the highest accuracy.

DEBATE Kim et al. (2024) introduce three roles: a Scorer providing an initial score and rationale, a Critic acting as a Devil's Advocate, and a Commander coordinating communication and memory. The iterative loop continues until the Critic accepts the judgment or a round limit is reached. Evaluated on SummEval (Fabbri et al., 2021) and Topical-Chat (Gopalakrishnan et al., 2023), DEBATE consistently outperforms automatic metrics (e.g., ROUGE, BERTScore) and prior LLM-as-a-judge systems (GPTScore, G-Eval, ChatEval). Performance improves with additional debate rounds and stronger Critic personas, though at increased inference cost.

ChatEval Chan et al. (2023) draw inspiration from collaborative human evaluation by assigning diverse roles to multiple agents and allowing them to debate before reaching a final verdict. Structured communication strategies (e.g., sequential vs. simultaneous debate) and position calibration are used to mitigate turn-order bias. Evaluated on open-ended QA (Zheng et al., 2023b; Wu et al., 2023) and dialogue generation (Gopalakrishnan et al., 2023; Mehri and Eskenazi, 2020), ChatEval demonstrates higher agreement with human judgments than single-agent evaluators and traditional automatic metrics, highlighting the promise of multi-agent debate for more reliable evaluation.

Dataset	Domain	Size	Fields
GSM8K	Math	250	`input`, `output`
OpenOrca	QA	250	`system_prompt`, `input`, `output`
MedRedQA	Medical	83	`document`, `input`, `output`

Table 2: Dataset statistics used in our experiments.

3 Datasets

Table 2 summarizes the datasets used in our evaluation. The selected datasets span three domains: (1) multi-step mathematical reasoning, (2) open-domain dialogue, and (3) specialized medical QA, allowing us to analyze the robustness of MADEF across distinct data distributions and task requirements. Due to computational constraints (each debate cycle required approximately two minutes per example), we randomly sampled 250 instances per dataset, except for MedRedQA, which contains only 83 available samples. The Fields column indicates which components were provided to the evaluator models during MADEF execution.

GSM8K Cobbe et al. (2021) consists of grade-school–level mathematical word problems requiring explicit multi-step reasoning. Unlike surface-level QA tasks, correctness depends on logical traceability and intermediate reasoning steps. This dataset allows us to examine whether MADEFs can distinguish between correct final answers and flawed—but well-written—reasoning chains.

OpenOrca Lian et al. (2023) provides diverse open-domain QA data distilled from large instruction-tuned models, including both factual and opinion-based queries. This dataset enables MADEF performance under unconstrained conversational settings where responses may be subjective, multi-intent, or stylistically varied.

MedRedQA Nguyen et al. (2023) contains expert-curated medical question-answer pairs sourced from clinical guidelines and peer-reviewed publications. Because responses require factual precision, safety-aware reasoning, and domain grounding, MedRedQA serves as a stress test whether MADEF can provide reliable judgments in high-risk, knowledge-sensitive applications.

4 Methods

4.1 Preparation

Data Preparation We construct paired evaluation data by prompting two LLMs: (1) Meta-Llama-3.1-8B-Instruct (Grattafiori et al., 2024) and (2) DeepSeek-R1-Distill-Llama-8B (DeepSeek-AI et al., 2025) to generate responses to the same set of inputs.[1] This setup allows each MADEF to compare outputs produced under identical conditions and ensures controlled evaluation across domains.

MADEF Replication We then apply the MADEF frameworks listed in Table 1 to evaluate and compare the generated responses. Each MADEF is reproduced using its default configuration, with minimal prompt and formatting adjustments required for compatibility with our datasets. This replication strategy enables a fair comparison across frameworks and avoids introducing methodological confounds. To establish a baseline, we implement a **single-agent evaluator**.

Model Scaling To accommodate resource constraints, we replace larger LLMs used in previous work (e.g., GPT-4 (OpenAI et al., 2024)) with smaller models, Gemma2:2B (Team et al., 2024). All MADEF agents are executed locally via `Ollama`, and the same evaluator model is used for both debating and judging roles. This allows us to examine whether relative performance trends of MADEFs hold under reduced computational budgets and reflect more realistic deployment environments.

Evaluation Rubrics Across all datasets, MADEFs assign scores on a 1–5 Likert scale according to their respective evaluation criteria, ensuring consistency and comparability across domains. The detailed criteria are provided in Table 4. We apply the following rubric focus:

- **GSM8K:** Logical validity and step-by-step reasoning for math problem-solving.
- **OpenOrca:** Robustness and bias reduction in open-ended, multidimensional dialogue.
- **MedRedQA:** Factual correctness and evidence-based assessment due to the high-stakes nature of the medical domain.

5 Results & Discussions

We summarize the performance of all MADEFs across datasets and analyze their scoring behavior, efficiency, and alignment with human judgments.

5.1 Scoring Tendencies & Variability

Table 3 reports the quantitative scores assigned by each MADEF across datasets and rubrics. Overall, we observe clear differences in scoring tendencies across frameworks. **Single-agent** evaluation consistently assigns the highest mean scores with very low standard deviation, indicating strong leniency and limited score variability. This suggests that single-agent systems are more prone to biased or overconfident judgments rather than discriminating between subtle response differences.

In contrast, **DEBATE** produces substantially lower mean scores and the largest variance across all datasets (e.g., GSM8K correctness: 2.584 ± 1.429), reflecting a more conservative and diverse scoring pattern. The wide dispersion indicates that adversarial critique leads to stronger differentiation among responses, but may also introduce instability when the evaluator model is weak or resource-constrained. **Debatable-Intelligence** and **MORE** fall between the extremes, showing moderate scores and variance, indicating balanced behavior. **ChatEval** shows consistently high but less inflated scores, while maintaining moderate variability, indicating selective but not overly permissive judgment.

Inference efficiency also varies significantly across frameworks. As shown in the Avg Time(s) rows, single-agent evaluation is the fastest (8–17s per example), followed by MORE and ChatEval. Debatable-Intelligence and DEBATE are the slowest, requiring up to 188s and 358s respectively, due to multi-round debate and role coordination. These results indicate that increased interaction depth comes at a substantial computational cost and may be impractical for large-scale deployment under limited resources.

5.2 Human Alignment & Domain-Specific Performance

Table 4 presents alignment between MADEF decisions and human judgments. No single MADEF dominates across all domains, directly addressing RQ1. In GSM8K, single-agent and MORE achieve the highest agreement, suggesting that lightweight frameworks can be sufficient for deterministic reasoning tasks. For OpenOrca, ChatEval achieves the highest agreement in completeness and helpfulness, reflecting its design for open-ended conversational settings. In MedRedQA,

[1] Detailed examples are provided in Table 6.

Dataset	Rubrics	Single Agent (Baseline)	Debatable-Intelligence	MORE	DEBATE	ChatEval
GSM8K	Correctness	4.940 ± 0.323 , 4.968 ± 0.308	4.984 ± 0.178 , 4.940 ± 0.323	4.996 ± 0.063 , 4.980 ± 0.140	2.584 ± 1.429 , 3.196 ± 0.861	4.804 ± 0.417 , 4.812 ± 0.411
	Reasoning	4.056 ± 0.291 , 4.300 ± 0.508	4.304 ± 0.460 , 4.032 ± 0.234	4.636 ± 0.481 , 4.472 ± 0.499	2.696 ± 1.346 , 3.196 ± 0.730	3.872 ± 0.675 , 4.164 ± 0.694
	Completeness	4.980 ± 0.166 , 4.984 ± 0.252	4.996 ± 0.063 , 4.968 ± 0.217	4.944 ± 0.247 , 4.628 ± 0.508	2.746 ± 0.996 , 3.066 ± 0.497	4.464 ± 0.580 , 4.536 ± 0.566
	Accuracy	4.979 ± 0.171 , 4.975 ± 0.289	4.992 ± 0.126 , 4.972 ± 0.208	4.952 ± 0.232 , 4.672 ± 0.486	2.832 ± 1.241 , 3.226 ± 0.542	4.760 ± 0.472 , 4.800 ± 0.438
	Avg time (s)	8.610	83.860	17.701	188.575	23.060
OpenOrca	Relevance	4.493 ± 0.782 , 4.804 ± 0.576	4.609 ± 0.664 , 4.232 ± 0.893	4.991 ± 0.094 , 4.272 ± 0.494	3.069 ± 0.419 , 3.214 ± 0.546	4.764 ± 0.487 , 4.804 ± 0.454
	Completeness	3.906 ± 0.819 , 4.188 ± 0.740	3.927 ± 0.657 , 3.668 ± 0.805	4.732 ± 0.444 , 3.932 ± 0.566	3.281 ± 0.526 , 3.406 ± 0.492	3.976 ± 0.658 , 4.364 ± 0.750
	Accuracy	4.061 ± 1.045 , 4.071 ± 0.842	3.907 ± 1.060 , 3.625 ± 1.107	4.920 ± 0.286 , 4.536 ± 0.553	3.053 ± 0.358 , 3.092 ± 0.432	4.605 ± 0.708 , 4.646 ± 0.556
	Clarity	4.232 ± 0.542 , 4.080 ± 0.713	4.000 ± 0.718 , 4.072 ± 0.539	4.948 ± 0.222 , 3.704 ± 0.626	3.297 ± 0.495 , 3.367 ± 0.482	3.944 ± 0.584 , 4.304 ± 0.772
	Helpfulness	4.080 ± 0.860 , 4.210 ± 0.855	3.972 ± 0.820 , 3.688 ± 0.876	4.956 ± 0.241 , 4.164 ± 0.491	3.317 ± 0.465 , 3.325 ± 0.468	4.168 ± 0.616 , 4.516 ± 0.701
	Avg time (s)	17.090	125.624	20.002	90.652	34.963
MedRedQA	Accuracy	3.554 ± 1.027 , 4.048 ± 0.582	3.573 ± 0.802 , 3.807 ± 0.930	4.855 ± 0.354 , 4.253 ± 0.537	2.897 ± 0.525 , 3.216 ± 0.406	4.115 ± 0.509 , 4.628 ± 0.627
	Appropriateness	4.313 ± 0.714 , 4.494 ± 0.571	4.451 ± 0.651 , 4.542 ± 0.630	4.265 ± 0.543 , 4.663 ± 0.590	3.359 ± 0.486 , 3.162 ± 0.371	3.923 ± 0.786 , 4.718 ± 0.532
	Safety	4.711 ± 0.553 , 4.675 ± 0.497	4.427 ± 0.609 , 4.904 ± 0.336	4.518 ± 0.592 , 4.590 ± 0.625	3.000 ± 0.000 , 3.014 ± 0.119	3.974 ± 0.603 , 4.551 ± 0.550
	Clarity	4.590 ± 0.585 , 4.831 ± 0.408	4.195 ± 0.922 , 4.735 ± 0.607	4.614 ± 0.514 , 3.855 ± 0.566	3.250 ± 0.437 , 3.143 ± 0.352	3.603 ± 0.744 , 4.513 ± 0.698
	Professionalism	4.614 ± 0.514 , 4.639 ± 0.508	4.512 ± 0.527 , 4.783 ± 0.415	4.963 ± 0.179 , 4.855 ± 0.270	3.193 ± 0.398 , 3.114 ± 0.320	4.205 ± 0.567 , 4.667 ± 0.501
	Avg time (s)	14.454	99.294	25.430	143.411	358.395

Table 3: Mean and standard deviation of model evaluation scores across GSM8K, MedRedQA, and OpenOrca. Values shown as Average ± Std for both LLaMA and Distill-LLaMA. Avg time (s) rows report the average inference time in seconds for each MADEF.

Dataset	Rubric	Definition	S	DI	M	D	CE
GSM8K	Correctness	Final answer correctness	**17**	16	**17**	13	16
	Reasoning	Logical, step-by-step derivation	**10**	4	7	6	9
	Completeness	No missing steps; reasoning fully shown	**19**	**19**	13	18	15
	Accuracy	Correct calculations and arithmetic	**9**	8	8	**9**	**9**
OpenOrca	Relevance	Addresses the question or task directly	8	7	**12**	0	4
	Completeness	Includes all necessary points and details	9	6	1	6	**16**
	Accuracy	Information correctness vs expected answer	11	**18**	16	4	10
	Clarity	Clear organization and understandable output	12	**16**	8	7	13
	Helpfulness	Practical usefulness for a typical user	15	**16**	12	6	6
MedRedQA	Medical Acc.	Clinical correctness and factual soundness	**17**	11	11	13	10
	Appropriateness	Suitability for patient context	15	7	**19**	2	16
	Safety	Avoidance of harmful or risky guidance	5	5	7	2	**12**
	Clarity	Understandable and well-structured explanation	**20**	15	1	14	12
	Professionalism	Ethical and respectful tone	**13**	10	10	6	10

Table 4: Evaluation rubrics, definitions, and agreement counts across automated methods. Bold indicates the highest score per row, and underline marks the second-highest. MADEF abbreviations: S = Single Agent, DI = Debatable-Intelligence, M = MORE, D = DEBATE, CE = ChatEval.

performance varies widely across rubrics, with MORE and ChatEval performing best on appropriateness and safety, while single-agent evaluation performs better on clarity and professionalism. These findings support **H1**: domain-specific characteristics influence which MADEF performs best. They also provide evidence for **H2**: increased complexity does not guarantee superior evaluation quality. Lightweight frameworks such as single-agent and MORE achieve competitive or even superior—alignment under certain conditions, suggesting that cost-effective configurations may be viable alternatives when resource constraints or scalability concerns apply.

5.3 Qualitative Behavioral Differences

Beyond aggregate scores, Table 5 provides a qualitative comparison of evaluator behavior across sampled examples. In GSM8K, all evaluators except DEBATE consistently converge on identical rankings, confirming that deterministic reasoning tasks require minimal debate interaction. DEBATE performs worst, producing unstable or null scores, supporting our earlier observation that increased interaction depth does not necessarily translate into better evaluation under constrained models.

In OpenQA, separation depends on task difficulty. For trivial factual items, all frameworks behave similarly. However, when responses require multi-part reasoning or factual discrimination, MORE and Debatable-Intelligence outperform others by correctly identifying subtle answer differences. ChatEval follows closely, single-agent evaluation is overly optimistic, and DEBATE shows the highest instability.

Differences sharpen in MedRedQA. MORE and ChatEval align most closely with human judgment by rewarding evidence and penalizing unsupported claims. Debatable-Intelligence performs reasonably well, while single-agent evaluation consistently overestimates quality. DEBATE is least reliable, often producing missing outputs.

Domain	Example	Ranking (Best→Worst)	Notes
GSM8K	Budget balance	MORE = DebInt = ChatEval = Single > DEBATE	All correct; DEBATE varies unnecessarily.
	Boat leakage	MORE = DebInt = ChatEval = Single > DEBATE	Stable reasoning; DEBATE least consistent.
	Money fraction	MORE = DebInt = ChatEval = Single > DEBATE	Simple arithmetic; debate adds no value.
	Trip duration	MORE = DebInt = ChatEval = Single > DEBATE	Consistent validation; DEBATE noisy.
	Award cost	MORE = DebInt = ChatEval = Single > DEBATE	Longer steps but correct; DEBATE unstable.
Overall	MORE (25) = DebInt (25) = ChatEval (25) = Single (25) > DEBATE (5)		Deterministic tasks expose DEBATE weaknesses.
MedRedQA	SSRI vs antihistamine	MORE ≈ ChatEval > DebInt > Single > DEBATE	Detects over-claims; Single too lenient.
	Bone density meds	MORE ≈ ChatEval > DebInt > Single > DEBATE	Evidence sensitivity benefits structured eval.
	COVID + smoking risk	MORE = DebInt = ChatEval > Single > DEBATE	Mostly correct; Single optimistic.
	Surgical risk evidence	MORE > DebInt ≈ ChatEval > Single > DEBATE	MORE penalizes missing evidence best.
	DLB hallucination case	MORE ≈ DebInt ≈ ChatEval ≈ Single > DEBATE	All reasonable; DEBATE erratic/null outputs.
Overall	MORE (25) > DebInt (19) = ChatEval (19) > Single (10) > DEBATE (5)		Evidence-heavy tasks need stricter judgment.
OpenQA	Detroit court authority	MORE ≈ DebInt ≈ ChatEval ≈ Single ≈ DEBATE	Trivial extraction; no separation.
	Duplicate question	MORE ≈ DebInt ≈ ChatEval ≈ Single ≈ DEBATE	Identical prompts; matches across evaluators.
	Multi-step story	MORE ≈ DebInt ≈ ChatEval > Single > DEBATE	Multi-part reasoning benefits multi-agent.
	Walking motivation	MORE ≈ DebInt > ChatEval > Single > DEBATE	Captures subtle reasoning differences.
	Documentary awards	MORE > ChatEval ≥ DebInt > Single > DEBATE	MORE separates hallucination vs accuracy.
Overall	MORE (25) > DebInt (19) > ChatEval (16) > Single (10) > DEBATE (5)		Open-ended tasks need finer discrimination.

Table 5: Qualitative ranking across sampled examples. Rankings reflect evaluator agreement on best-to-worst judgments (5=best, 1=worst).

Overall, these qualitative findings reinforce our quantitative conclusions: (1) no single MADEF dominates across all domains, (2) lightweight frameworks can match or exceed complex ones under certain task conditions, and (3) debate-heavy designs may be counterproductive when evaluator models lack sufficient reasoning stability.

6 Conclusion and Future Work

In this work, we present a systematic comparison of four Multi-Agent Debate Evaluation Frameworks (MADEFs) across three domains: mathematical reasoning, open-domain dialogue, and medical question answering. Our analysis shows that although all MADEFs can approximate human judgment, they differ substantially in scoring behavior, stability, and computational efficiency. These findings clarify that greater debate depth or interaction complexity does not inherently yield higher-quality evaluation, and that lightweight configurations can perform competitively when extensive cross-agent reasoning is unnecessary.

Our results further reveal meaningful differences in where each framework is most effective: deterministic reasoning tasks benefit little from multi-agent interaction, whereas evidence-sensitive and open-ended settings gain from structured critique and role diversity. By establishing a standardized experimental setup and a unified single-agent baseline, this study provides a foundation for more principled comparisons of emerging LLM-based evaluation approaches.

Looking ahead, we will extend our evaluation pipeline to support variable debate depth, adaptive role assignment, and more flexible scoring strategies. We also plan to examine how MADEF behaviors evolve under larger models and heterogeneous agent configurations, as well as explore additional reliability signals such as cross-framework agreement and self-consistency. Ultimately, our goal is to enable more scalable, transparent, and resource-aware evaluation paradigms that broaden the practical applicability of LLM-as-a-judge systems.

7 Limitations

Although our study provides a controlled comparison of MADEFs, several limitations remain. First, all evaluators, including both MADEF agents and the single-agent baseline, were implemented using a single underlying model (Gemma 2B), which limits diversity in reasoning styles and may underestimate the benefits of debate under higher-capacity or heterogeneous agents. Second, most evaluations were conducted without gold-standard reference labels, restricting our ability to assess absolute correctness and requiring reliance on relative agreement rather than ground-truth accuracy. Third, our setup fixes debate depth, role configurations, and stopping conditions, preventing analysis of how MADEF behavior changes under adaptive interaction settings. Finally, prompt design was intentionally kept minimal to ensure comparability across frameworks; more expressive prompting or rubric refinement may yield different outcomes and potentially improve single-agent performance.

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A Example from the dataset

Table 6 illustrates the structure and content of typical examples used in the MADE evaluation. This format demonstrates the diversity of input types and ground truth answers, the varying complexity found in model-generated outputs, and how auxiliary information was incorporated for the different domains.

Dataset	input	output	llama_output	distill_llama_output	extra(system_prompt or document)
GSM8K	Janet's ducks lay 16 eggs per day. She eats three for breakfast every morning and bakes muffins for her friends every day with four. She sells the remainder at the farmers' market daily for $2 per fresh duck egg. How much in dollars does she make every day at the farmers' market?	Janet sells 16 - 3 - 4 = «16-3-4=9»9 duck eggs a day. She makes 9 * 2 = $«9*2=18»18 every day at the farmer's market. ### 18 ..... Janet sells 16 - 3 - 4 = «16-3-4=9»9 duck eggs a day. She makes 9 * 2 = $«9*2=18»18 every day at the farmer's market. ### 18	To find out how much Janet makes every day at the farmers' market, we need to determine the number of eggs she sells. First, let's calculate the total number of eggs she lays per day: Janet's ducks [...] eggs left to sell per day: 16 - 7 = 9. Since she sells each egg for $2, we can calculate her daily earnings: Eggs sold per day: 9. Price per egg: $2. Daily earnings: 9 * $2 = $18. ### [final answer] $18	First, I need to determine how many eggs Janet lays each day. She lays 16 eggs per day. Next, I'll calculate the number of eggs she uses for her muffins. She uses 4 eggs daily for baking muffins. Th … **Eggs Sold at the Farmers' Market:** [12 eggs] … **Total Money Made per Day:** [12 eggs] × $2 per egg = $24. ### [final answer] \boxed{24}	N/A
OpenOrca	Answer the following question: Given the background: Middle adulthood lasts from the mid-30s to the mid-60s. During this stage of life, many people raise a family and strive to attain career goals. Th … He labeled the first group as group A; and the latter group as group B. Answer the following question: Would group B have faster or slower reaction time than group A?	It is generally accepted that reaction time tends to decline with age, so it is likely that group B, who are in their twenties, would have faster reaction times than group A, who are in middle adulthood … Overall, it is reasonable to assume that group B would have faster reaction times on average than group A due to the effects of aging on neural processing speed and physical changes.	To address David's report, we need to consider the physical changes that occur in middle adulthood, as mentioned in the background. One of the notable changes is the decline in physical abilities, inc … This difference is due to the natural decline in physical abilities that occurs with aging, including changes in the nervous system, brain processing speed, and muscle mass and strength.	Okay, so I'm trying to figure out whether Group B, which is in their twenties, would have faster or slower reaction times compared to Group A, who are in middle adulthood, like their mid-30s to mid-60s … Group A may experience slower reaction times due to aging, decreased strength, and potential health issues affecting the nervous system. Therefore, Group B is expected to have faster reaction times.	system_prompt: You are an AI assistant. You will be given a task. You must generate a detailed and long answer.
MedRedQA	- 29 - Male - USA I'm very confused why different labs have different ranges. For example, for WBC, these are the "normal" ranges by lab – LabCorp 3.4–10.8 ([This example report] ...) … which is normal for their ranges. But that's still considered low on other websites. Does anyone know why this is? And if I need to be concerned or not about my "normal" results that seem to be low?	Nothing to be worried about. Depending upon the exact method and equipment and reagents used, every lab has to calibrate its tests, and decide upon what's "normal" for their lab (e.g. the values seen … different stores – they are all different sizes even though they are supposed to be the same; or like buying the same color of paint from different paint manufacturers – the exact tint is different).	The discrepancy in white blood cell (WBC) count ranges between different laboratories is due to various factors. Here are some possible reasons: 1. **Methodology and calibration:** Different laboratories … Remember, a low WBC count can be a sign of an underlying issue, but it's not always a cause for concern. Your doctor will help determine the best course of action based on your individual situation.	Okay, so I'm trying to figure out why different labs have different normal ranges for things like WBC count. I'm a bit confused because when I had my labs done, my WBC came out as 3.7 at LabCorp, whic … If you're not having symptoms associated with low WBC (e.g., fatigue), there is typically no cause for concern. It's advisable to discuss this with a healthcare provider to assess any underlying factors or trends in your results.	document: The Current Status of Telemedicine Technology Use Across the World Health Organization European Region: An Overview of Systematic Reviews Background Several systematic reviews evaluating … Trial Registration PROSPERO (International Prospective Register of Systematic Reviews) CRD42022309375; https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=309375

Table 6: Examples from the datasets. *Input* corresponds to the question, *Output* is the gold label, *llama_output* is generated by `Meta-Llama-3.1-8B-Instruct`, and *distill_llama_output* is generated by `DeepSeek-R1-Distill-LLaMA-8B`. *OpenOrca* includes an additional field *system_prompt*, and *MedRedQA* includes *document*. Texts are truncated with an ellipsis (. . .) when they exceed the available space.